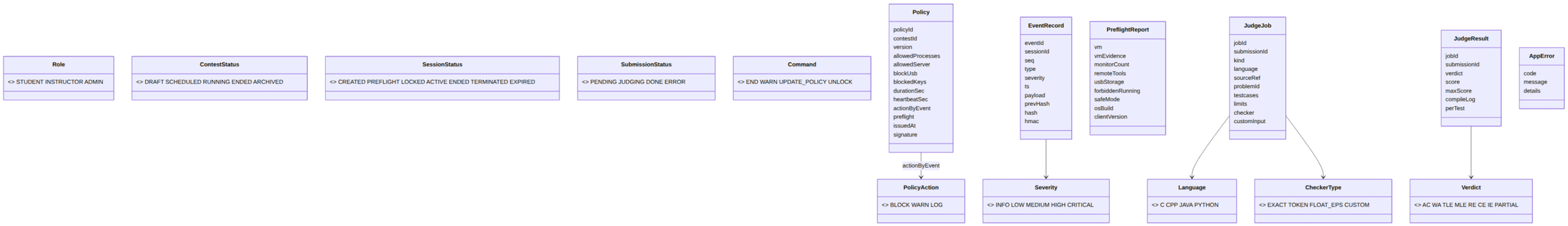
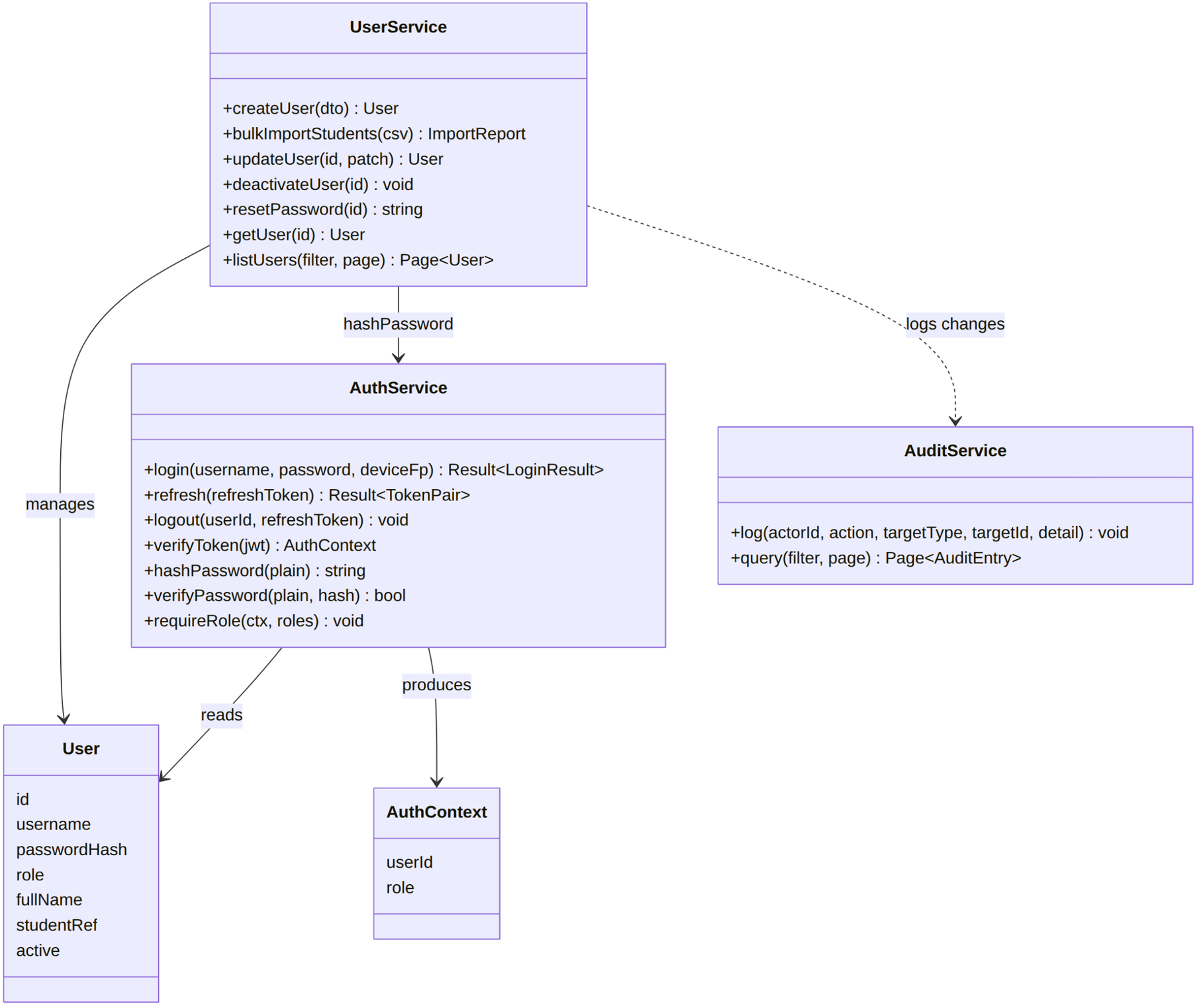
1. Shared types (enums and DTOs)

LLD section 1 | Class diagram 1 of 10



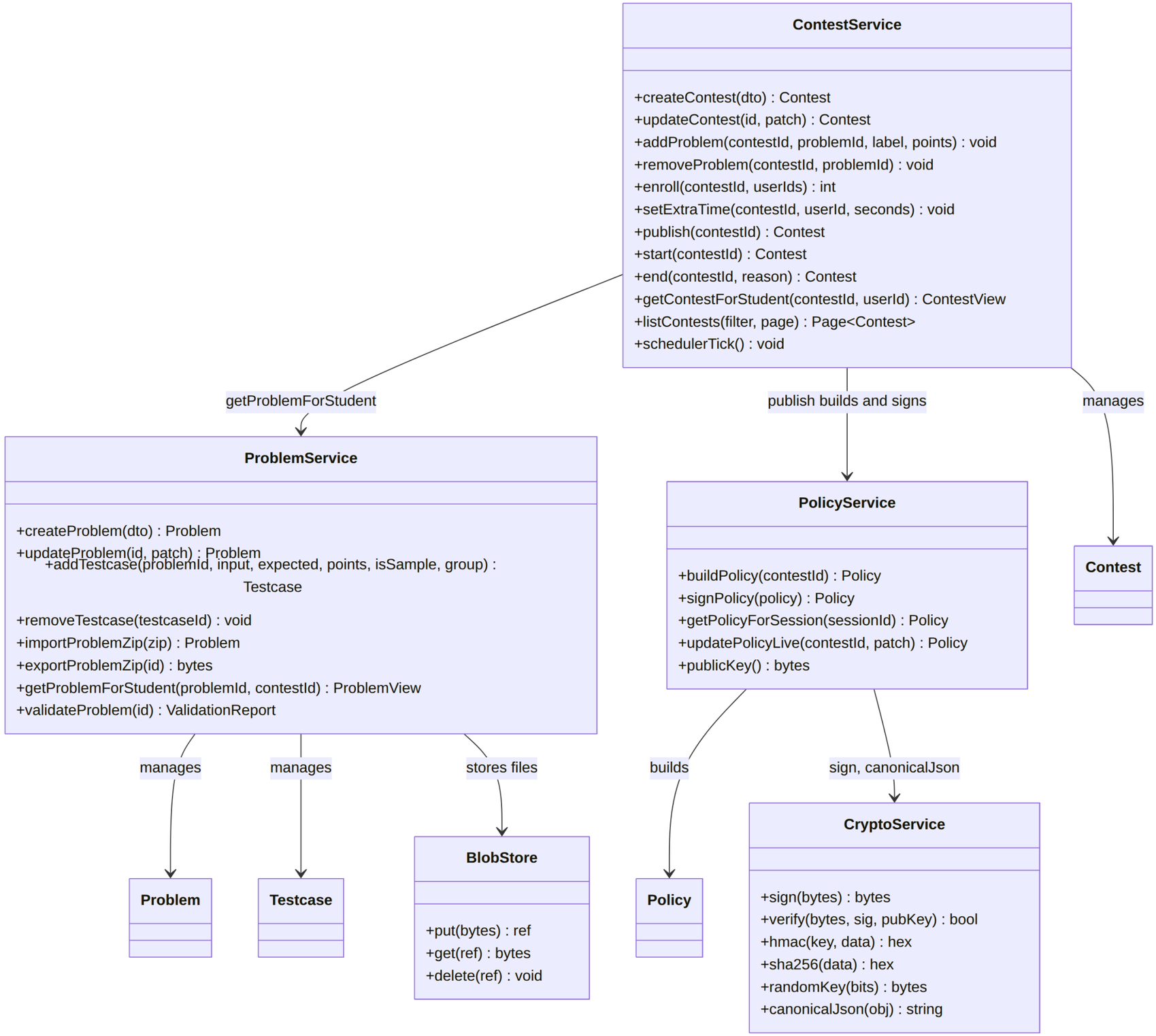
2. Server: Auth and Users

LLD section 3.1, 3.2, 3.16 | Class diagram 2 of 10



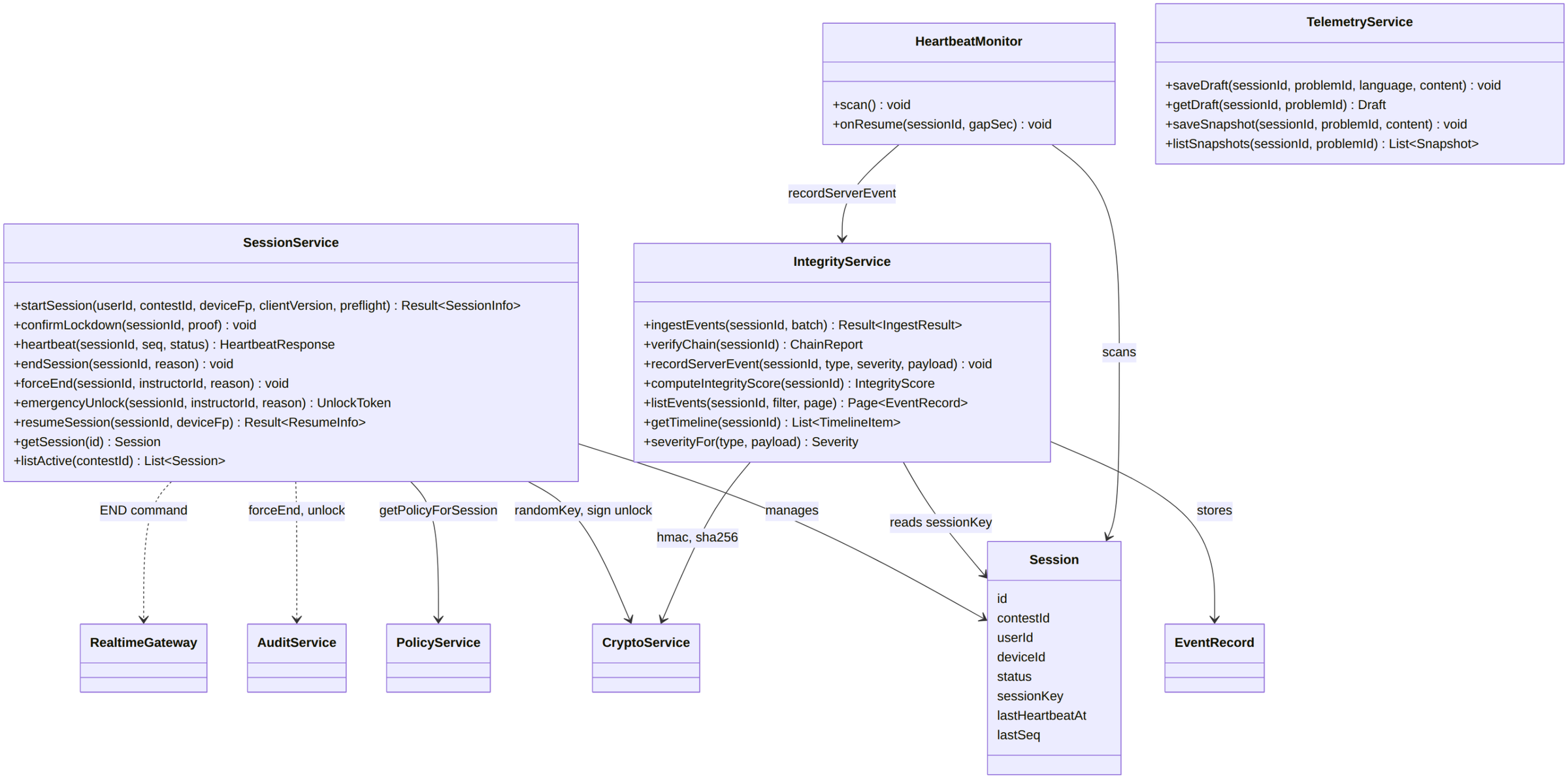
3. Server: Problems, Contests, Policy

LLD section 3.3 to 3.5, 3.18 | Class diagram 3 of 10



4. Server: Sessions, Events, Heartbeat, Telemetry

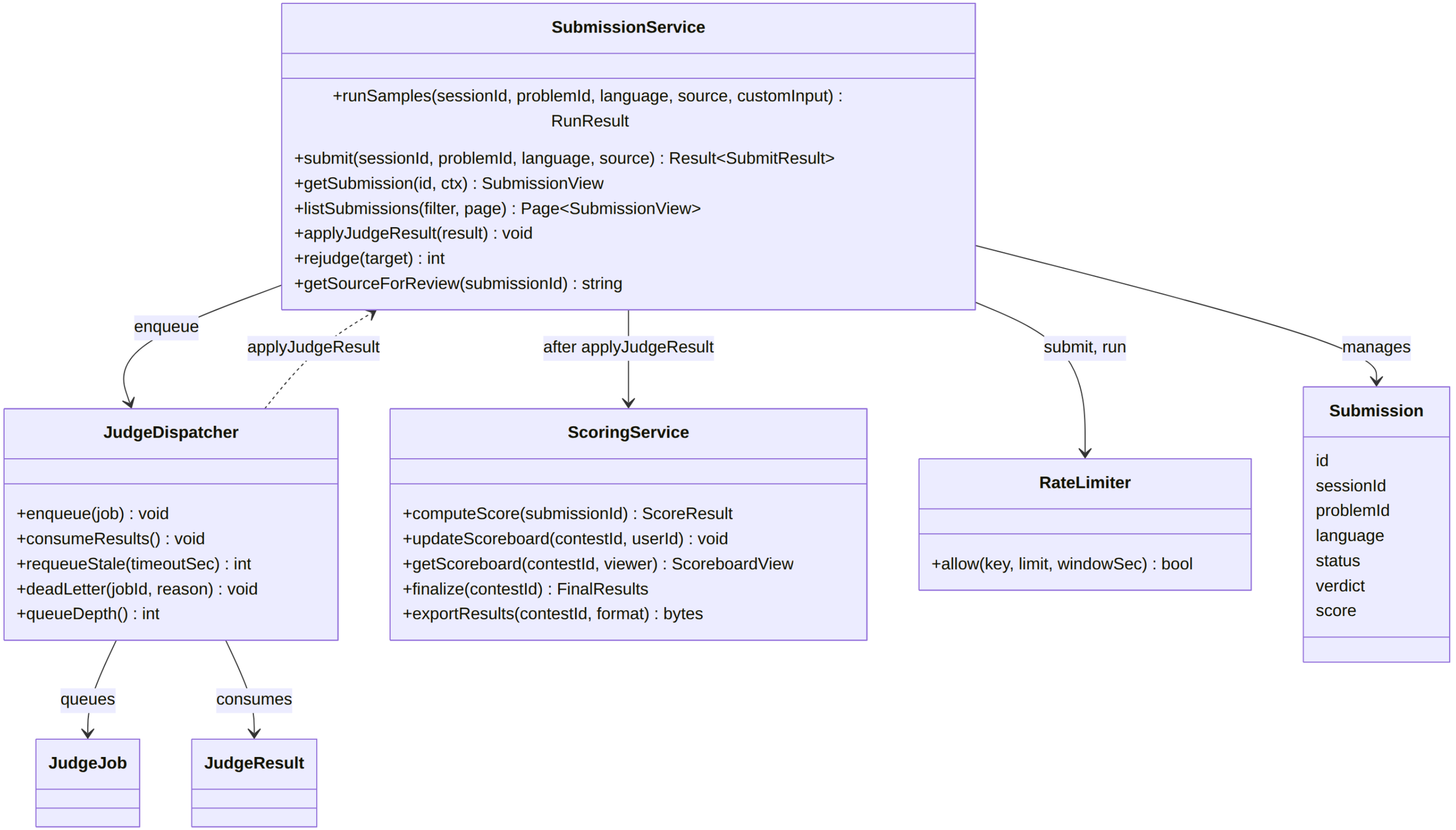
LLD section 3.6, 3.10 to 3.12 | Class diagram 4 of 10



Design note: sessionKey must be recoverable (encrypted at rest or derived from a server master key). A session_key_hash alone cannot verify HMACs.

5. Server: Submissions, Judge dispatch, Scoring

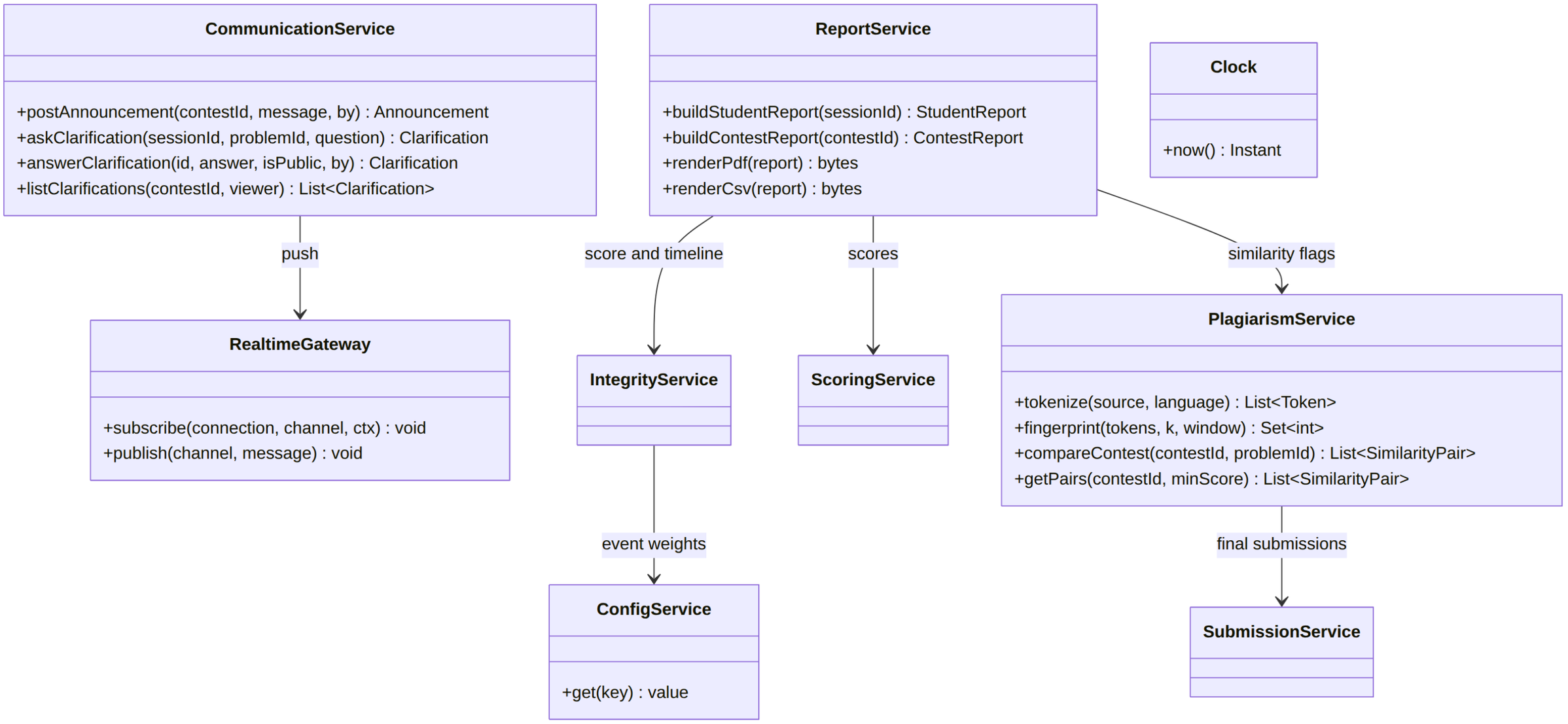
LLD section 3.7 to 3.9 | Class diagram 5 of 10



Week-1 shortcut: replace JudgeDispatcher with JudgeAdapter.runAll(...) calling Judge0 directly (no Redis, no queue).

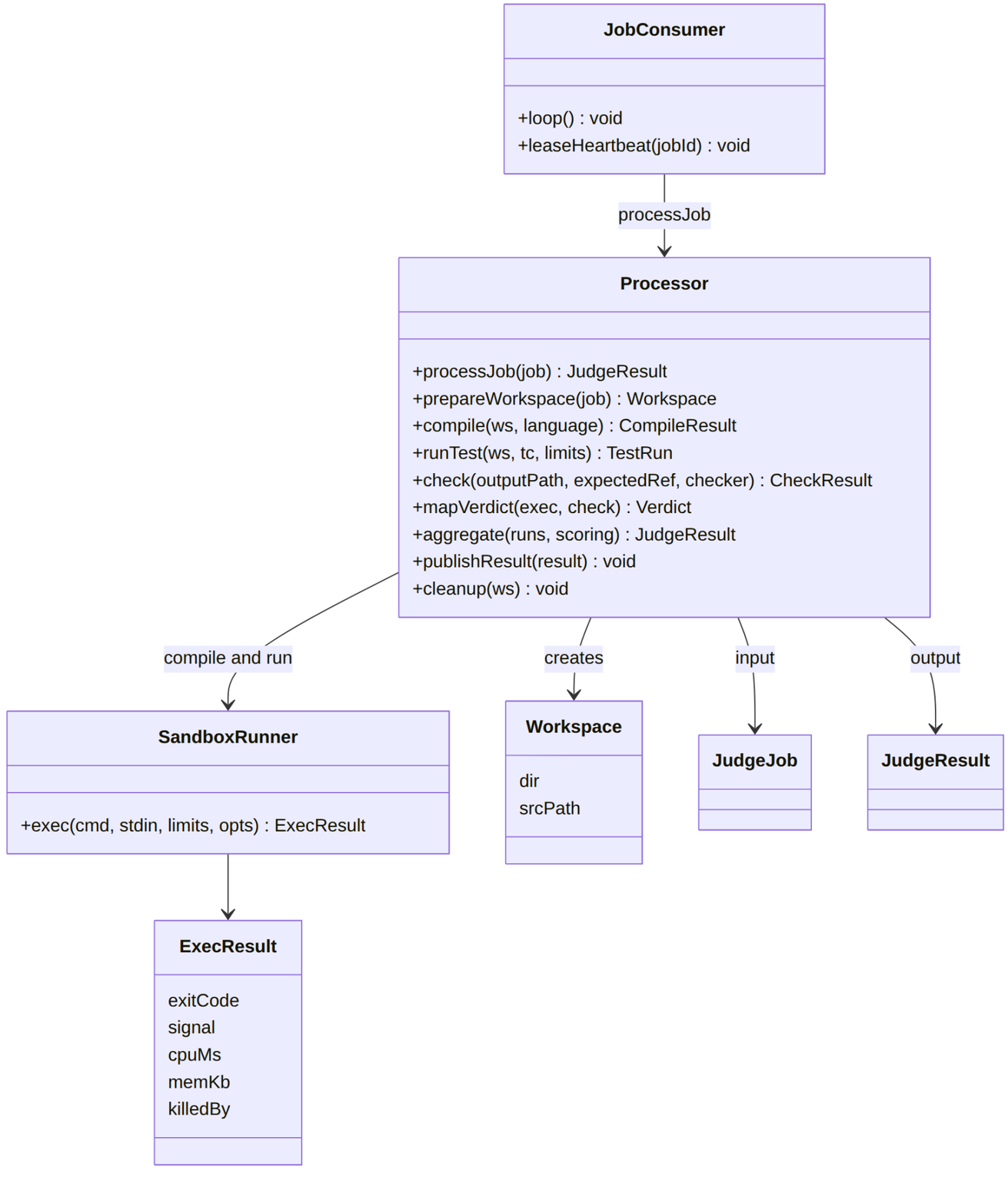
6. Server: Reports, Plagiarism, Communication, Realtime, Utilities

LLD section 3.13 to 3.18 | Class diagram 6 of 10



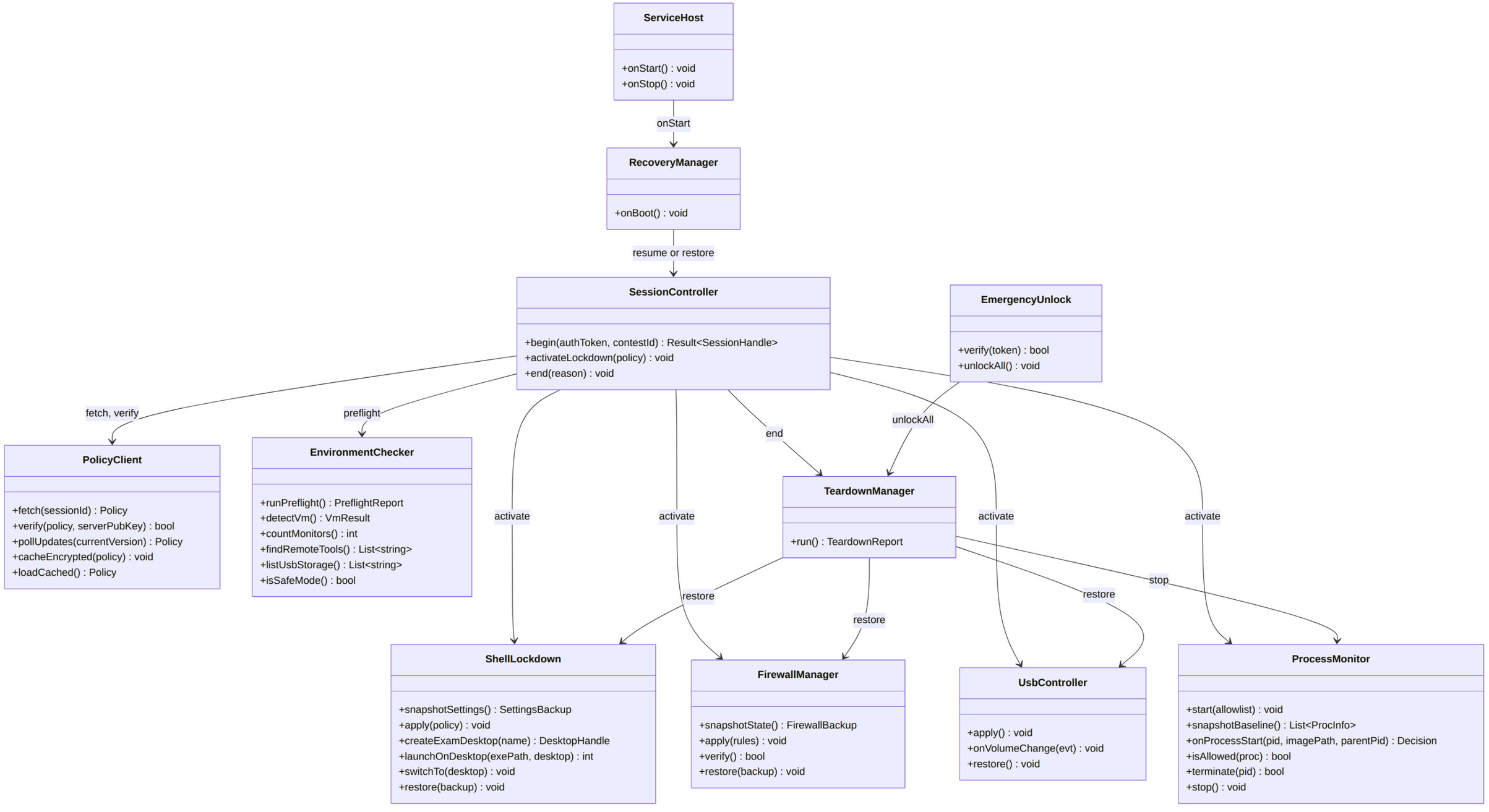
7. Judge Worker (separate host, no DB access)

LLD section 4 | Class diagram 7 of 10



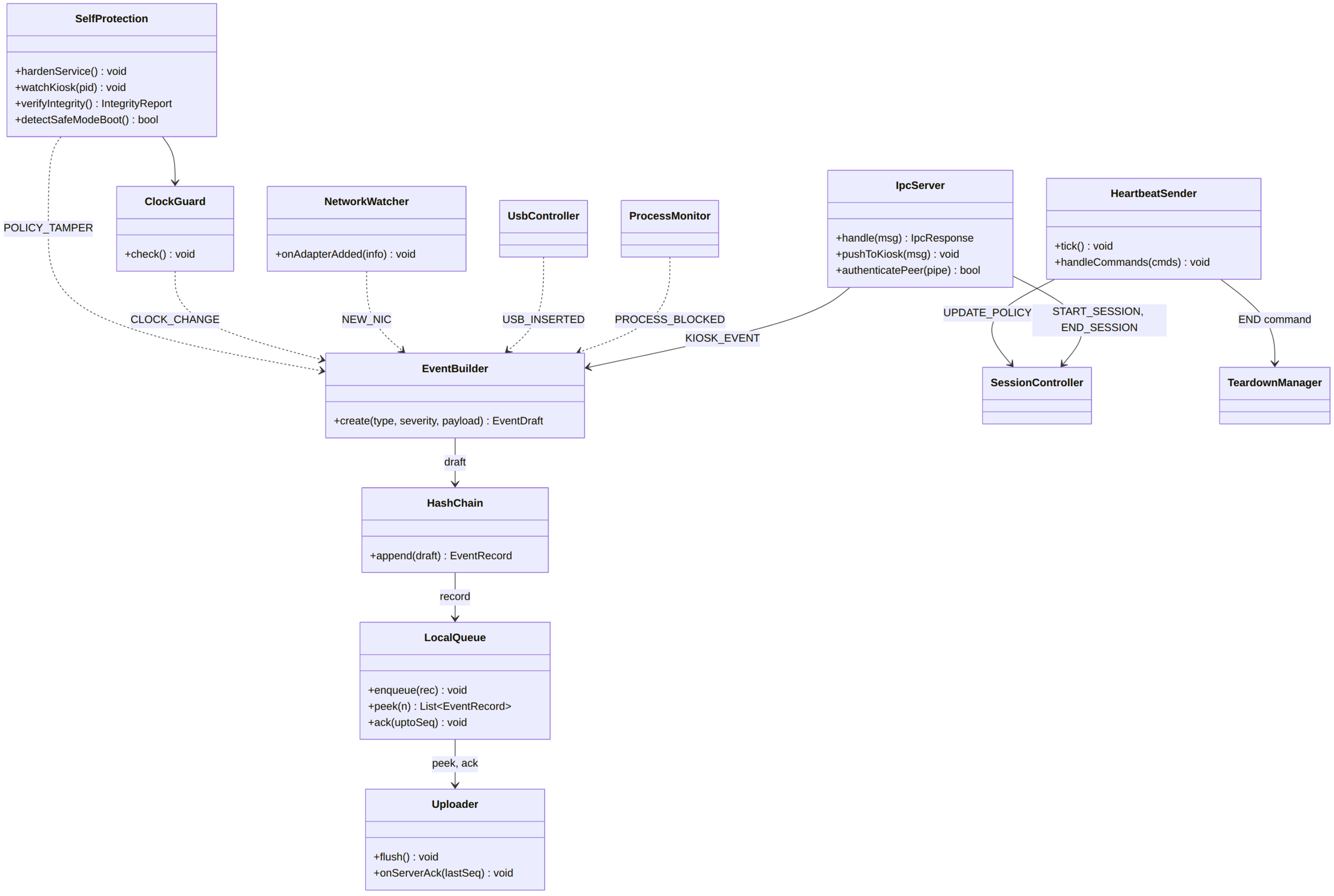
8. Watchdog: Session and Lockdown modules

LLD section 5.1 to 5.9, 5.12 | Class diagram 8 of 10



9. Watchdog: IPC, Event pipeline, Self-protection

LLD section 5.2, 5.10, 5.11 | Class diagram 9 of 10



10. Kiosk app and Web frontend

LLD section 6, 7 | Class diagram 10 of 10

